

Simple Linear Regression Model Inference

Module 3

Overview

- Consider making inferences *only after* verifying the linear regression assumptions
- We are interested in inference for $\hat{\beta}_1$ (Why?)
- Recall:

Scalar Form

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i \quad \text{where} \quad \epsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

$$\hat{\beta}_1 = \frac{SS_{XY}}{SS_{XX}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = r_{xy} \frac{s_Y}{s_X}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$s^2 = MSE = \frac{\text{SSE}}{\text{degrees of freedom for error}} \\ = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - 2} = \frac{\sum_{i=1}^n e^2}{n - 2}$$

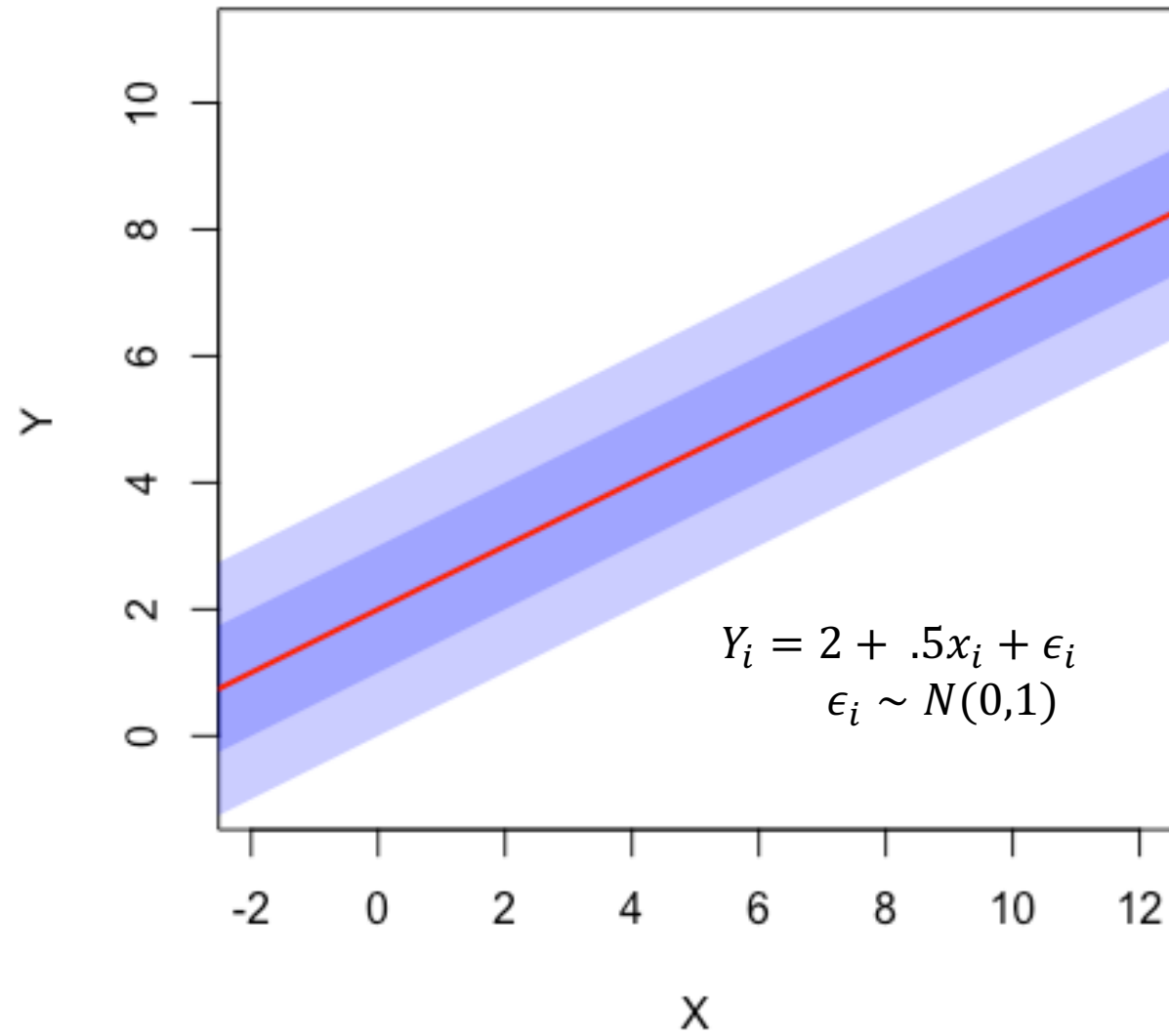
Matrix Form

$$Y = X\beta + \epsilon \quad \text{where} \quad \text{Var}(\epsilon) = \sigma^2 I$$

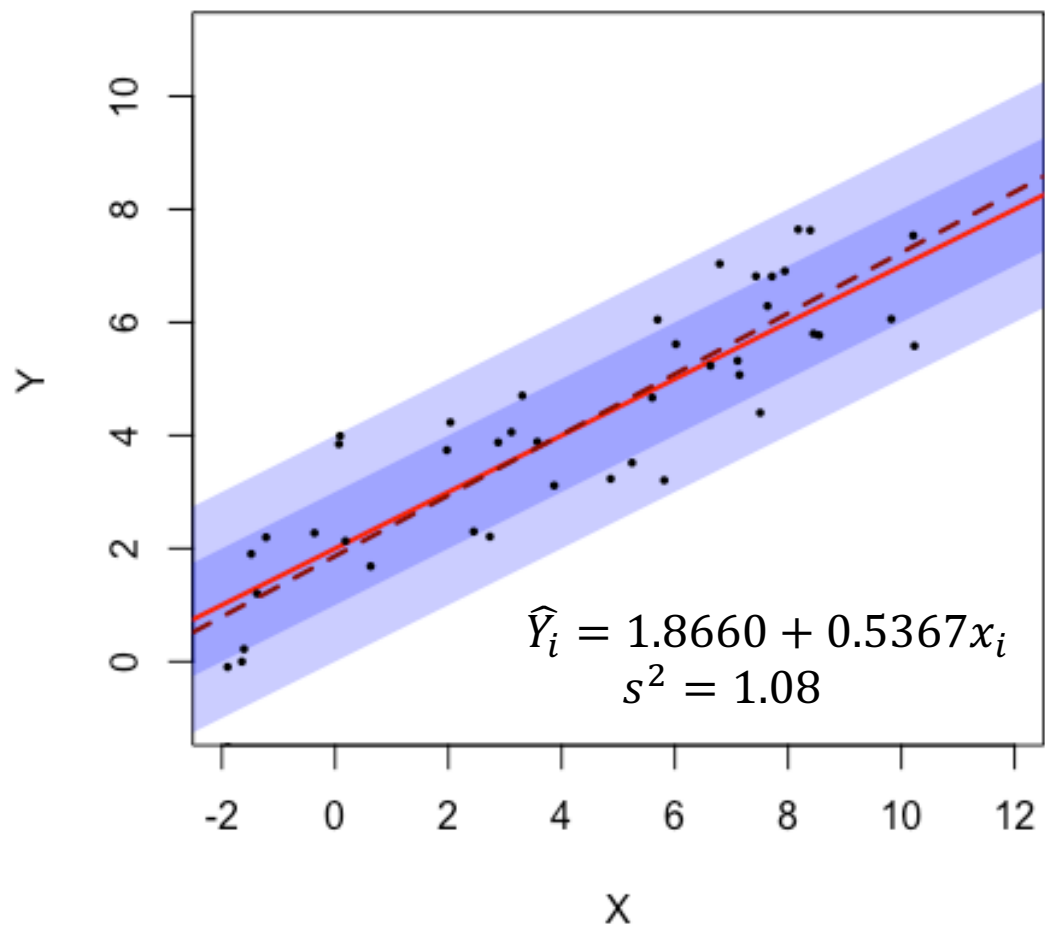
$$\hat{\beta} = (X'X)^{-1}X'Y$$

$$s^2 = MSE = \frac{e'e}{n - 2}$$

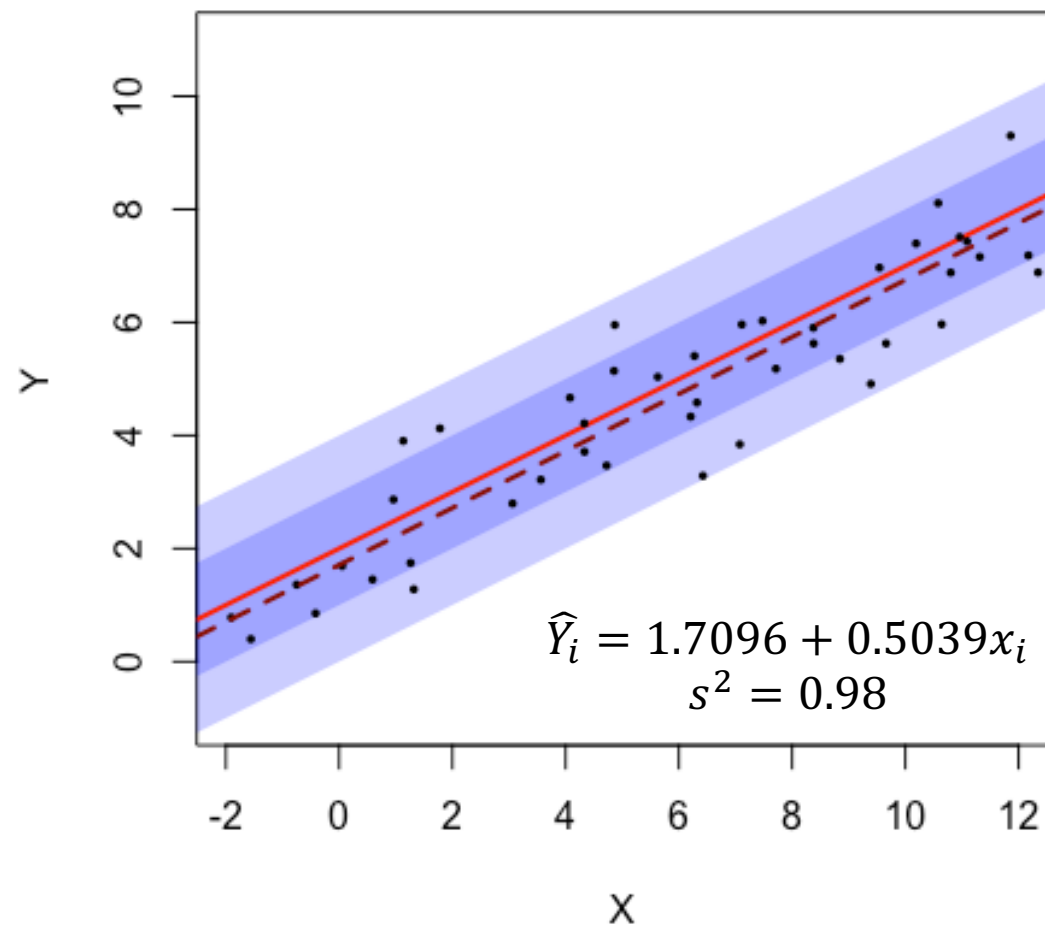
Population (Truth)



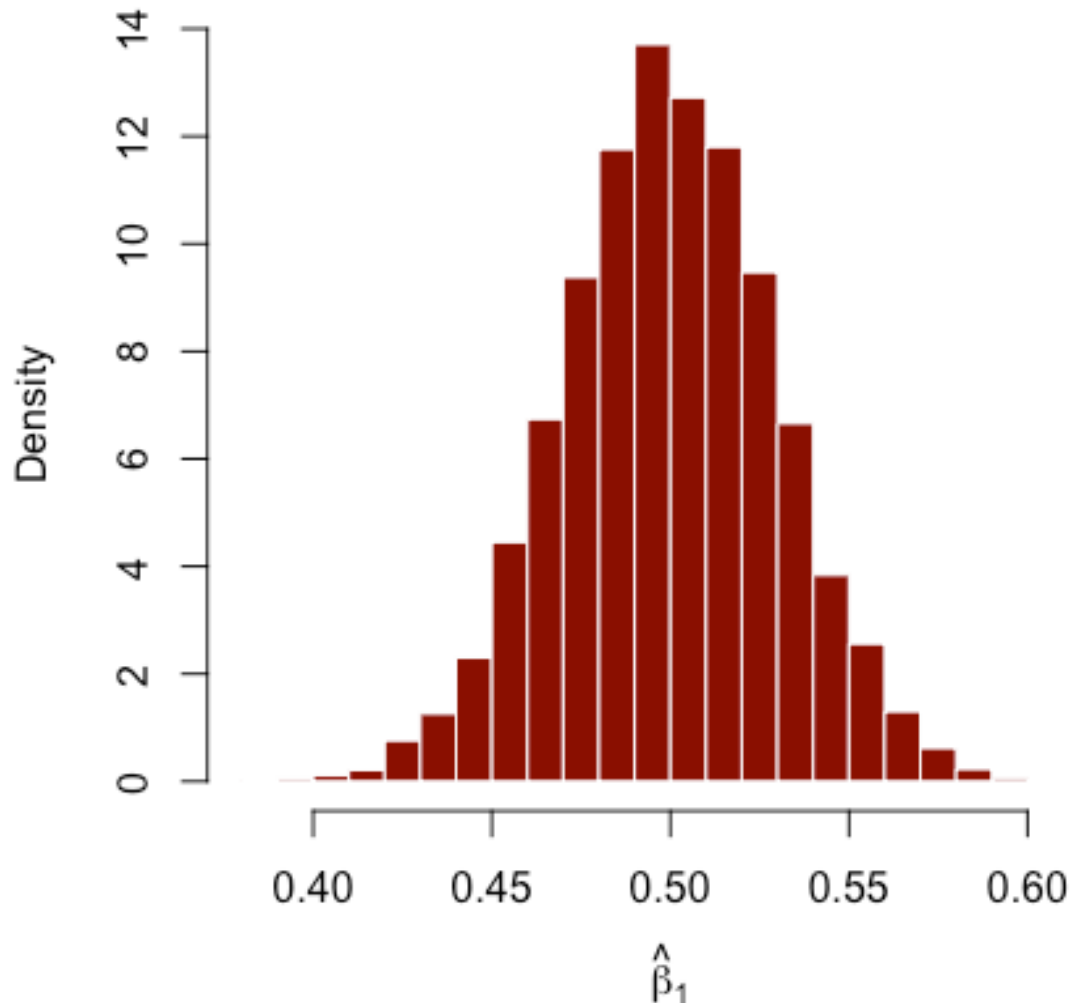
Sample of size 50 (Estimate of Truth)



Different Sample of size 50 (Estimate of Truth)



Histogram of $\hat{\beta}_1$ for 10,000 samples



This histogram represents an **approximation** of the **sampling distribution of $\hat{\beta}_1$** (for samples of size 50)

Given the true regression equation

$$Y_i = 2 + .5x_i + \epsilon_i$$
$$\epsilon_i \sim N(0,1)$$

Can we actually derive the **true** sampling distribution?

Distribution of $\hat{\beta}_1$

$$\begin{aligned}
 \hat{\beta} &= (X'X)^{-1}X'Y \\
 &= (X'X)^{-1}X'(X\beta + \epsilon) \\
 &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\epsilon \\
 &= \beta + (X'X)^{-1}X'\epsilon
 \end{aligned}$$

$$\begin{aligned}
 E[\hat{\beta}] &= E[\beta + (X'X)^{-1}X'\epsilon] \\
 &= \beta + (X'X)^{-1}X'E[\epsilon] \\
 &= \beta + (X'X)^{-1}X'(0) \\
 &= \beta
 \end{aligned}$$

$$\begin{aligned}
 \text{Var}[\hat{\beta}] &= \text{Var}[\beta + (X'X)^{-1}X'\epsilon] \\
 &= (X'X)^{-1}X' \text{Var}[\epsilon]((X'X)^{-1}X')' \\
 &= (X'X)^{-1}X' \sigma^2 I X (X'X)^{-1} \\
 &= \sigma^2 (X'X)^{-1}X'X(X'X)^{-1} \\
 &= \sigma^2 (X'X)^{-1}
 \end{aligned}$$

So,

$$\hat{\beta} \sim N(\beta, \sigma^2 (X'X)^{-1})$$

$$\hat{\beta} \sim N(\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$$

The standard deviation of $\hat{\beta}_1$,
in scalar form, is given by

$$\text{s.d.}(\hat{\beta}_1) = \sqrt{[\sigma^2(\mathbf{X}'\mathbf{X})^{-1}]_{22}}$$

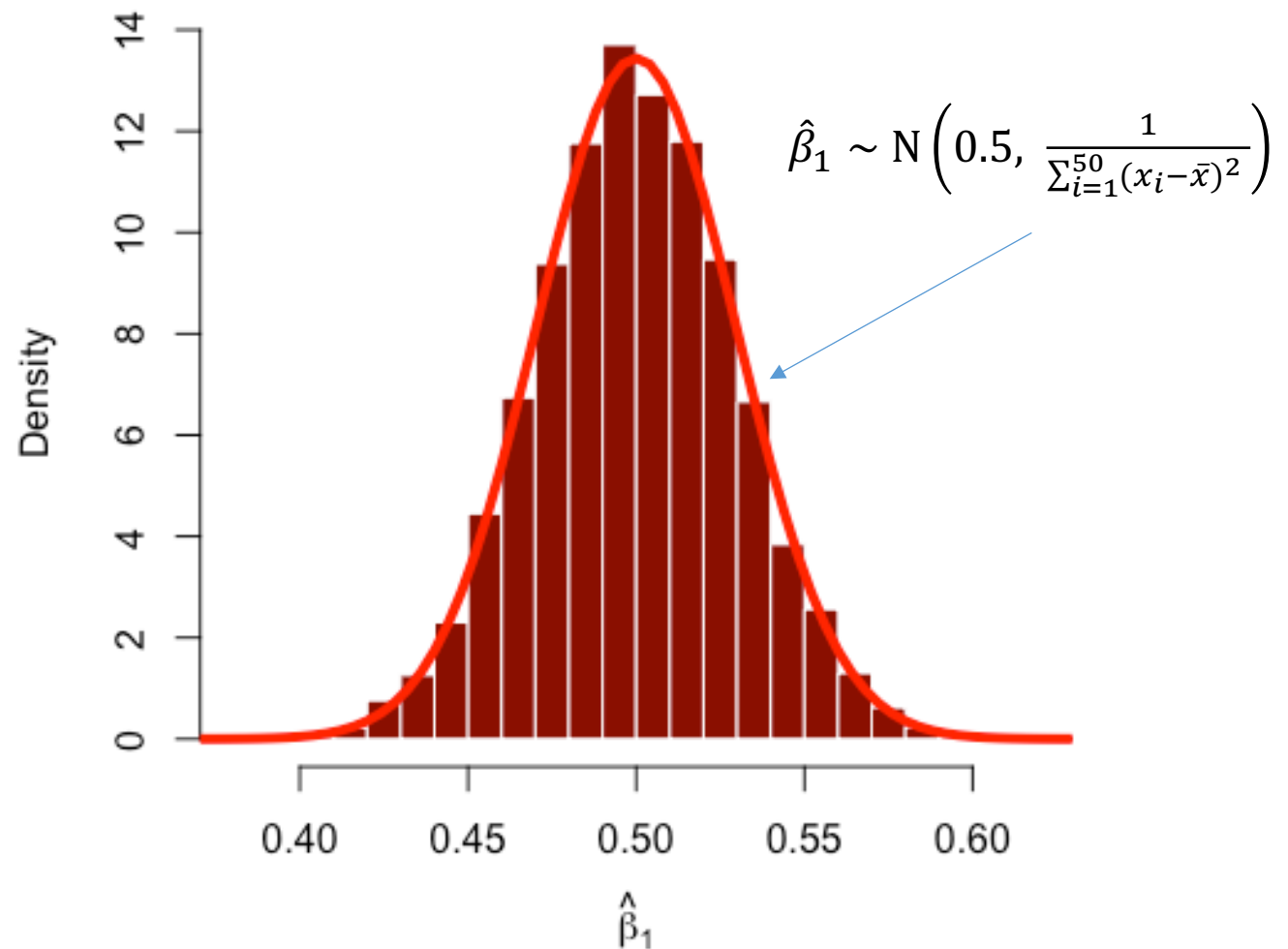
$$\text{Recall: } (\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{SS_{XX}} \begin{bmatrix} \frac{1}{n} \sum_{i=1}^n x_i^2 & -\bar{x} \\ -\bar{x} & 1 \end{bmatrix}$$

So,

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}\right)$$

Great! This sampling distribution gives us a way to quantify uncertainty about our sample estimates of β_1 .

Histogram of $\hat{\beta}_1$ for 10,000 samples



Estimating the Distribution of $\hat{\beta}_1$

Problem: we still don't know β_1 and σ^2 .

- BUT, we do have estimates for β_1 and σ^2 ($\hat{\beta}_1$ and s_e^2). We will use these to approximate the sampling distribution, from which we will make inference (confidence intervals and hypothesis tests)
- We will standardize (center to 0 and divide by the **standard error**) before proceeding. Let

$$T = \frac{\hat{\beta}_1 - \beta_1}{s_e / \sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}} = \frac{\hat{\beta}_1 - \beta_1}{s_e / \sqrt{S_{XX}}} = \frac{\hat{\beta}_1 - \beta_1}{\text{std. err.}(\hat{\beta}_1)}$$

- It can be shown (when the assumptions hold) that T has a t -distribution with $n - 2$ degrees of freedom, that is

$$\frac{\hat{\beta}_1 - \beta_1}{\text{std. err.}(\hat{\beta}_1)} \sim t_{n-2}$$

*** Notice that the degrees of freedom is:
sample size – number of mean parameters estimated.

Confidence Interval for the Slope

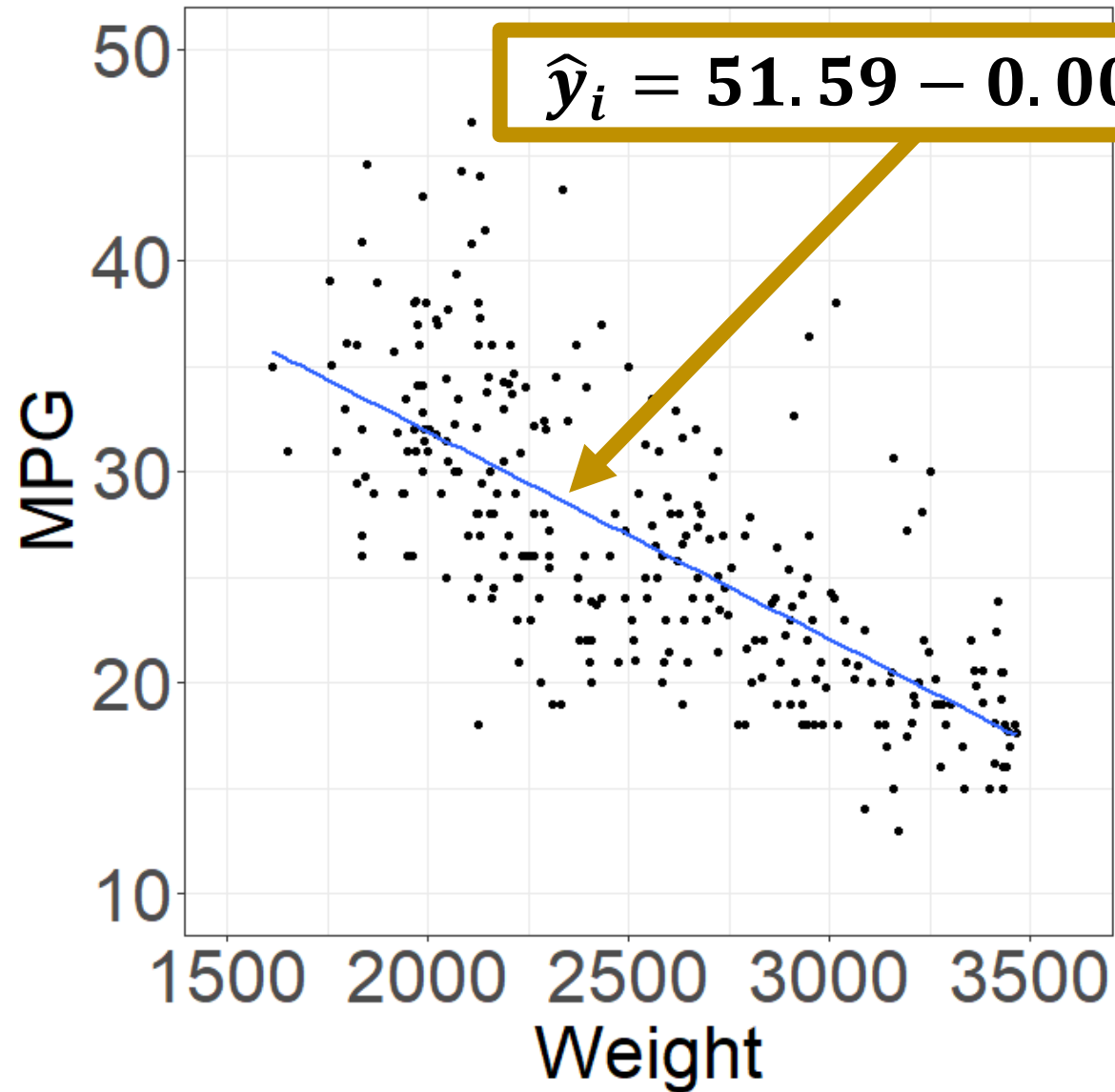
- Given the results on the previous slide, a $(1 - \alpha)100\%$ confidence interval for β_1 is given by

$$\hat{\beta}_1 \pm t_{\alpha/2} \widehat{std.err.}(\hat{\beta}_1),$$

where $t_{\alpha/2}$ represents the upper $\alpha/2$ critical value from the t distribution with $n - 2$ degrees of freedom and

$$\widehat{std.err.}(\hat{\beta}_1) = s_e / \sqrt{S_{XX}}$$

Car Gas Mileage



$$n = 289$$

$$s_e^2 = 22.31$$

$$\widehat{std. err.}(\hat{\beta}_1) = \frac{s_e}{\sqrt{SS_{XX}}} = \frac{s_e}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

A 95% confidence interval for the slope is given by

$$\hat{\beta}_1 \pm t_{\alpha/2} \widehat{std. err.}(\hat{\beta}_1),$$

Hypothesis Test for the Slope

- To test $H_0: \beta_1 = 0$, versus $H_a: \beta_1 \neq 0$, the t statistic is given by

$$t_{obs} = \frac{\hat{\beta}_1 - 0}{\widehat{std.err.}(\hat{\beta}_1)}$$

- The p -value is computed as $p = 2P(t > |t_{obs}|)$, where t_{obs} is the observed value of the test statistic, and the tail probability is based on a t distribution with $n - 2$ degrees of freedom.
- If we use a significance level of α , then we reject H_0 if $p < \alpha$. In other words, this indicates evidence at the α -level that there is a significant linear relationship between x and y .
- The default test in the `lm()` output in R is for testing $H_0: \beta_1 = 0$, versus $H_a: \beta_1 \neq 0$.

Car Gas Mileage

- For the car gas mileage data, the test statistic to test $H_0: \beta_1 = 0$, versus $H_a: \beta_1 \neq 0$ is given by

$$t_{obs} = \frac{\hat{\beta}_1 - 0}{\widehat{std.err.}(\hat{\beta}_1)} = \frac{-0.00983}{0.00057} = -17.106$$

- The p -value is given by $p = 2P(t > |-17.106|)$, based on a t distribution with 287 degrees of freedom.

$$p = \text{essentially } 0$$

p-Values

- Recall that a p-value is the probability of seeing a result as “extreme” or more than what you observed if H_0 is true.
- In a hypothesis test for the slope, the p-value is the probability of having a slope as “steep” as the one our data produced, if the “truth” is that there is no linear association between x and y .
- Remember:
 - if p-value $< \alpha$, we “reject” the null hypothesis in favor of the alternative hypothesis
 - if p-value $> \alpha$, we “fail to reject” the null hypothesis – we do not “accept” the null hypothesis, we simply have insufficient evidence to reject the null hypothesis

Confidence Intervals and Hypothesis Testing

- Two approaches to statistical inference: **confidence intervals** and **hypothesis testing**, will *generally* result in the same substantive conclusions.
- If a confidence interval does not contain the null value, then the hypothesis test will result in a “significant” p-value, and vice-versa.
- Often, a confidence interval is more useful for assessing the magnitude of an effect than a single p-value.

Note: we could follow the same procedure and get a confidence interval and do a hypothesis test for the intercept $\hat{\beta}_0$, but this is usually less interesting and unnecessary to the research question.